CS5228 DATA MINING PROJECT

Predicting HDB monthly rent in Singapore

CS5228 project progress report

October 2026

Internal out-of-time validation; no Kaggle test score reported

Task, data and evaluation question

Target Monthly approved rent (SGD) for an individual HDB rental record.

Labeled train 150,000 rows, Jan 2021–Mar 2025.

Unlabeled test 50,000 rows, Mar 2025–Jul 2026; 49,620 postdate the final train month.

Project questions

- What do property, time and location explain?
- Do blocks and amenities improve prediction?
- Do COE and stocks add a reliable leading signal?

Most test rows are future-dated: validation must move forward in time.

Source: `reports/train_test_eda/summary.json`; CS5228 project brief (provided)

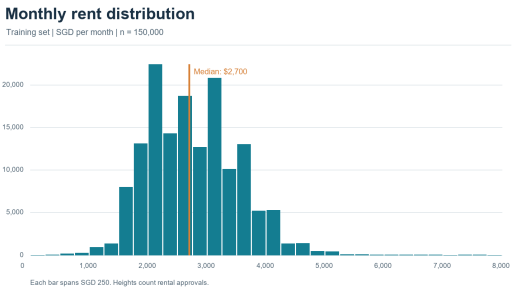
Data audit: clean core table, but important caveats

Check	Observed result and treatment
Missing/duplicates	No empty core cells; 425 exact train duplicates. Retain ambiguous copies because there is no rental ID to prove error.
Constant fields	`FURNISHED` is always yes; `FEE` is always zero. Exclude from learning.
Inconsistency	One lease-commencement year exceeds approval year; flag the resulting age rather than assuming it is valid.
Categories	Normalize labels such as `4 room` / `4-room`; fit categorical encoding on training months only.
Auxiliary sentinels	Treat block `NIL` and `-1` as missing, not physical values.

The stock file also has missing quote fields; its data quality is separate from the complete rental table.

Source: `reports/train_test_eda/summary.json; reports/auxiliary_data_eda/summary.json`

Property EDA: rent spread, size and flat type



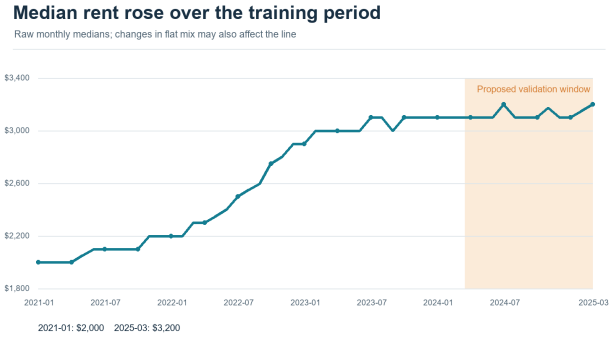
Flat type	*n*	Median SGD
2-room	3,098	2,100
3-room	49,024	2,350
4-room	54,079	2,800
5-room	35,523	3,000
Executive	8,234	3,000

Area: 31–215 m^2; raw area–rent $r = 0.329$.
One-room: just 42 rows.

Rent: min 300; Q1 2,100; median 2,700; Q3 3,200; P95 4,000; P99 4,600; max 7,600 SGD. Raw associations can reflect other features.

Source: `reports/train_test_eda/summary.json`; `reports/train_test_eda/01_rent_distribution.png`

Temporal EDA: a strong nonstationary level



Raw monthly median rent rose from SGD 2,000 (Jan 2021) to SGD 3,200 (Mar 2025). This is a 60% level increase, not necessarily a same-flat appreciation estimate.

Month, recency and a forward holdout belong in the baseline.

Source: `reports/train_test_eda/summary.json`; `reports/train_test_eda/02_monthly_median.png`

Spatial EDA: town heterogeneity and block coverage

Selected town	*n*	Median SGD
Bukit Timah	438	3,000
Central	2,278	3,000
Bishan	3,306	2,900
Punggol	6,668	2,850
Bedok	8,907	2,500
Yishun	7,913	2,500

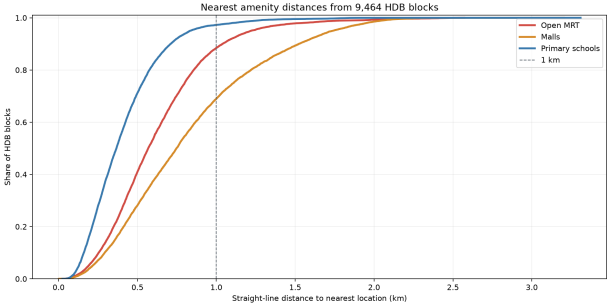
Across 26 towns, median rent ranges from SGD 2,500 to 3,000. Town sizes and flat mixes differ.

Normalized town–block–street keys match **100%** of train and test rows to 9,464 block records.

Coordinates, planning area, subzone, completion year and maximum floor add within-town detail.

Source: `reports/train_test_eda/summary.json; reports/auxiliary_data_eda/summary.json`

Amenity EDA: accessible, but snapshot-based



Nearest	Median km	$\leq$1 km
MRT	0.568	88.46%
Mall	0.747	68.97%
Primary school	0.366	97.23%

192 unique open MRT locations, 113 malls and 180 unique primary-school locations feed straight-line distance and count features.

Facility files are current snapshots: historical opening/closure dates are not available, so temporal interpretation is limited.

Source: `reports/auxiliary_data_eda/summary.json`; `reports/auxiliary_data_eda/02_nearest_distances.png`

Economic associations have not yielded stable row-level gains

Pair of monthly series	Transformation	Pearson r
Raw mean rent vs COE C premium	Levels; COE lag 1	$+0.960$
Mix-adjusted rent vs COE C	Growth; COE lag 1	$+0.306$
Mix-adjusted rent vs SG stock	Growth/return; lag 1	-0.181

Growth correlations use about 37 prevalidation months. The mix-adjusted rent index controls flat type/model, town, area and lease start; its adjustment is fitted before Apr 2024.

Against the same E baseline, adding engineered COE and SG-stock features together worsened row-level RMSE in all four forward windows by SGD 0.31–3.84. Individual economic sets sometimes helped a window slightly, but not consistently.

Monthly association is not evidence of a stable forward prediction gain.

Source: `reports/temporal_correlations/pair_correlations.csv`; `reports/economic_feature_ablation/forward_scores.csv`

Grouped time series: descriptive signal, mixed forecasts

Planning-area correlations

Lagged COE C bids-to-quota ratio vs area rent growth, across 24 eligible areas:

Rent statistic	Positive	Median r
Mean	20/24	+0.148
Median	17/24	+0.089

Correlation is modest; neither sign nor magnitude establishes a causal effect.

Pooled panel Ridge, median-rent RMSE

Forward start	+COE	+stocks
Jan 2023	+22.63	+16.78
Jul 2023	+9.78	+1.24
Apr 2024	−17.50	−2.76

Deltas are SGD vs history-only; positive is worse. This one-step setup uses the *observed* prior-month rent, unlike batch test forecasting.

Source: `reports/group_temporal_panel/report.md`; `reports/group_temporal_panel/group_correlations.csv`

Baseline protocol and leakage controls

- **Forward split:** fit Jan 2021–Mar 2024 (114,842 rows); hold out Apr 2024–Mar 2025 (35,158 rows).
- **Target and metrics:** record-level rent in SGD; compare RMSE, MAE, mean prediction bias, and month-level errors.
- **Seven core features:** flat type, model, area, age, approval month index, month-of-year, town.
- **Preprocessing:** normalize labels; learn imputation, scaling and categorical encoding from fit months only.
- **Ablation:** add block, amenity, COE and stock groups on identical rows, split and model settings.
- **Forecasting caveat:** one-month-updated signals differ from a fixed-origin forecast for all future test months.

Source: `reports/model_baseline/results.json; reports/model_families/results.json; reports/auxiliary_ablation/results.json`

Baselines: feature groups change the answer

Model / cumulative feature group	RMSE	MAE	Bias
Fit-period mean, no features	846.75	690.60	−531.44
CatBoost A: type, model, area	761.42	647.22	−534.06
CatBoost B: A + time/age	529.96	397.86	−49.91
CatBoost C: B + town	**487.22**	**364.09**	−48.29

All scores are Apr 2024–Mar 2025 row-level SGD errors. CatBoost: 400 iterations, depth 6, learning rate 0.05, $L_2 = 3$, seed 42. No tuning on the test file.

Time and age are added together in B; the A-to-B gain is not time's isolated contribution.

Source: `reports/model_baseline/model_metrics.csv; reports/model_baseline/results.json`

Model-family check on the same seven core features

Model	RMSE	MAE	Bias
Ridge regression ($\alpha = 10$)	624.88	468.18	+361.99
Random forest (160 trees, depth 18)	493.77	370.19	−56.65
CatBoost (400 trees, depth 6)	487.22	364.09	−48.29
LightGBM (400 trees, 31 leaves)	**485.09**	361.41	−42.68

Same 35,158-row holdout and seven features, with model-appropriate preprocessing. Ridge has a strong positive mean bias; trees handle nonlinear property/location effects better here.

LightGBM leads CatBoost by only SGD 2.12 RMSE on this single split.

Source: `reports/model_families/family_metrics.csv`; `reports/model_families/results.json`

Auxiliary ablation: blocks and amenities help most

HistGradientBoosting feature set	Features	RMSE	Bias
A: property	3	758.51	−531.85
B: + time/age	6	527.21	−48.35
C: + town	7	485.72	−46.78
D: + block attributes	15	480.22	−46.90
E: + amenities	29	**478.82**	−46.16
F: + COE	41	480.66	−62.72
G: + stocks	47	481.31	−65.93
H: + interactions	50	481.58	−71.83

Same model (200 boosting iterations, 31 leaves, learning rate 0.05) and holdout throughout. E improves over C by SGD 6.89 RMSE (1.42%). F/G test the original macro feature sets; both worsen this split. A separate engineered-signal comparison also finds no consistent forward gain. Do not compare this ablation to LightGBM as a pure feature effect.

Source: `reports/auxiliary_ablation/ablation_metrics.csv; reports/economic_feature_ablation/forward_scores.csv`

Diagnostics: useful signals, persistent drift

E: block + amenities

RMSE 478.82; MAE 356.36; mean bias −46.16 SGD. In Mar 2025, RMSE 457.45 but bias −80.10: average underprediction persists.

CatBoost C has RMSE 594.62 on 1,778 executive-flat holdout rows; only 14 one-room rows are too sparse for a stable slice.

E permutation importance

Largest MSE increases in a 2,000-row validation sample: flat type 151k, subzone 100k, area 31.6k; nearest mall 2.5k and MRT 2.3k.

Correlated features can share importance; this is predictive reliance, not an amenity price effect.

More forward splits and month-/group-level error audits matter more than a tiny single-split gain.

Source: `reports/auxiliary_ablation/feature_importance.csv`; `reports/auxiliary_ablation/monthly_errors.csv`; `reports/model_baseline/flat_type_errors.csv`

Current conclusion and next experiments

Current evidence

Property, time/age, block and amenity features improve the current row-level baseline. Some economic indicators show monthly associations and a late grouped one-step gain, but the tested economic feature sets have not delivered a stable row-level forward improvement.

Next experiments

1. Keep E as the feature benchmark; test leading model families on the same expanded set.
2. Repeat fixed-origin splits at a horizon near the future test span; test recency and group trends without future observed rent.
3. Revisit only availability-safe economic lags if they improve those tests; audit historical amenity coverage and sparse groups before final refit.

Source: `reports/auxiliary_ablation/results.json`; `reports/group_temporal_panel/report.md`; `reports/economic_feature_ablation/forward_scores.csv`